EpiKG: End-to-End Epistemic Preservation in Clinical Knowledge Graphs for Assertion-Aware Retrieval-Augmented Generation

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Abstract

Clinical NLP extracts assertion status (negated, uncertain, family-history, etc.), but these labels are routinely discarded before retrieval, causing downstream QA to treat epistemically distinct facts as equivalent—a failure we call the *epistemic propagation gap*. We show that closing this gap requires two components, not one: preserving assertion metadata end-to-end *and* routing retrieval by question intent. Assertion metadata alone does not help—adding it without routing yields no improvement (-3.2 pp, C3→C4, $p = 0.37$); the entire gain is unlocked by intent-aware routing ($+22.2$ pp, C4→C4g, $p < 10^{-10}$).

We validate this principle in EpiKG, a clinical KG-RAG system evaluated on ClinicalBench (400 questions, 43 MIMIC-IV patients, 9 assertion-sensitive categories). With Claude Opus 4.6, intent-aware KG-RAG reaches 69.0% versus 21.8% for LLM-alone ($+47.2$ pp; 95% BCa CI: [+40.9 pp, +52.8 pp]). Structured retrieval contributes $+28.2$ pp (C1→C3); routing contributes $+22.2$ pp (C4→C4g). Long-context concatenation (C6) reaches only 39.0%, failing completely on historical and sequence categories (0% each).

Cross-model validation on four models from distinct training lineages (Opus 4.6, GPT-OSS 20B, MedGemma 27B, Qwen3.5 35B) shows a consistent pattern: models with weaker baselines gain more from KG-RAG, and all four converge to 57.5–69.0% under C4g despite a 17 ppbaseline spread—suggesting the epistemic KG provides a knowledge floor that partially compensates for model-specific weaknesses. In a blinded physician pilot ($n = 30$), C4g answers were rated more correct (81% vs. 29%), safer (88% vs. 50%), and more helpful (81% vs. 29%) than LLM-alone.

1 Introduction

A physician documents "patient denies chest pain, sister had MI at 45, will consider statin if lipids remain elevated"—a single sentence encoding four epistemic states: a negated symptom, a family-history event, a hypothetical action, and an implicit present condition. Clinical NLP can detect these assertions accurately [1, 2], yet when a RAG system ingests the same note, assertion context is flattened: negation is lost, family attribution collapses onto the patient, and downstream reasoning may conclude the patient *has* chest pain and *had* an MI. We call this the *epistemic propagation gap*—the systematic loss of assertion status as clinical information moves from extraction through storage to retrieval.

The gap persists because, to our knowledge, prior KG-RAG systems do not propagate assertion labels beyond the extraction stage into graph-augmented retrieval. Clinical NLP pipelines (cTAKES, John Snow Labs) detect assertions, but do not carry them as edge attributes into knowledge graph construction or retrieval. OMOP excludes negated conditions from `CONDITION_OCCURRENCE` [3]; FHIR provides `verificationStatus` for `Condition` resources only, covering a fraction of the clinical data model. Graph-augmented RAG systems—GraphRAG [4], GFM-RAG [5], KARE [6], Medical-Graph-RAG [7]—build KGs that discard the metadata distinguishing "patient has diabetes" from "rule out diabetes." Similarly, recent clinical QA systems—MedPaLM 2 [8], Med-Gemini [9], AMIE [10]—excel at medical knowledge recall but are not designed for longitudinal assertion reasoning over real patient records, which is the gap ClinicalBench targets. A parallel *temporal inte-*

gration gap exists: clinical events span three time dimensions (valid time, transaction time, NLP-asserted temporality), yet existing systems model at most two [11, 12].

We present EpiKG, a clinical KG-RAG system that preserves epistemic status end-to-end, and ClinicalBench, a 400-question benchmark that exposes the propagation gap. Our contributions are primarily *empirical findings and a design principle*, not system components:

1. **The epistemic propagation gap.** We identify and formalize the systematic loss of assertion metadata across clinical NLP pipelines, derive an information-theoretic loss bound (Section 3.3, Appendix B), and implement end-to-end preservation as a testable invariant.

2. **Assertions alone do not help; routing is the active ingredient.** Adding assertion metadata to graph edges without question-type-specific retrieval does not improve accuracy (-3.2 pp, C3→C4; CI crossing zero, $p = 0.37$). The entire C3→C4g gain ($+19.0$ pp) is attributable to intent-aware routing ($+22.2$ pp, C4→C4g; $p < 10^{-10}$). Metadata enrichment without retrieval alignment is insufficient.

3. **Knowledge-compensator pattern.** Across four models from distinct training lineages, models with weaker baselines gain more from KG-RAG, and all converge to 57.5–69.0% under C4g despite C1 baselines spanning 20.5% to 37.0%. This convergence pattern—observed descriptively across four data points—is consistent with the epistemic KG providing a knowledge floor that partially compensates for model-specific weaknesses.

4. **ClinicalBench.** A 400-question benchmark across 9 assertion-sensitive categories (negation, uncertainty, conditional, family history, sequence, current state, duration, historical, change) that exposes category×condition interactions invisible to aggregate metrics. Released with Croissant metadata on HuggingFace.

5. **A falsifiable design principle.** Preserve epistemic metadata *and* route retrieval by question intent, then evaluate by category×condition interaction rather than aggregate score. This is not a system architecture but a falsifiable constraint: any pipeline violating it should underperform on assertion-sensitive longitudinal QA.

Our central research question is: *under what conditions does structured epistemic context help, hurt, or break even relative to unstructured retrieval and no retrieval?* The answer is interactional rather than uniform: gains depend on question type, routing policy, and model. Intent-aware KG-RAG improves all nine ClinicalBench categories, with largest effects on cross-admission longitudinal tasks; assertions without routing do not improve temporal categories; and brute-force long context does not recover these failures despite a large token window.

2 Related Work

We organize prior work along four axes and synthesize the gaps that EpiKG addresses (Table 9, Appendix).

Medical RAG systems. Graph-augmented retrieval has emerged as a leading paradigm for medical QA. GraphRAG [4] introduced community-based summarization; GFM-RAG [5] trained an 8M-parameter graph foundation model across 60 KGs (14M+ triples); KARE [6] adapted community retrieval to clinical decision support; DoctorRAG [13] emulates physician reasoning; MedRAG [14] constructs hierarchical diagnostic KGs; Medical-Graph-RAG [7] links documents via triple graphs. Concurrent work addresses EHR-specific retrieval: EHR-RAG [15] evaluates retrieval strategies over structured EHR tables, and MediGRAF [16] builds medical KGs via multi-agent extraction. Yet no published medical RAG system models assertion status or temporal context; GFM-RAG and KARE build population-level graphs rather than patient-level graphs from clinical text, and a recent survey [17] confirms no system incorporates epistemic metadata into the retrieval graph.

Clinical QA benchmarks and systems. Large-scale medical QA systems have advanced rapidly: MedPaLM 2 [8] reaches 86.5% on MedQA-USMLE via ensemble refinement; Med-Gemini [9] extends to multimodal clinical reasoning; AMIE [10] achieves specialist-level diagnostic dialogue. These systems target *medical knowledge recall*—answering exam-style questions from parametric knowledge or medical literature. ClinicalBench targets a complementary task: *assertion-faithful longitudinal reasoning* over real EHR records, where the challenge is not knowing that metformin treats diabetes but knowing whether *this patient* is still on metformin, whether a condition was ruled out or confirmed, and how the clinical picture changed across admissions. EHR-specific QA benchmarks exist—emrQA [18] (1M+ question-answer pairs from clinical notes) and EHR-RAG [15] (structured EHR retrieval)—but neither evaluates assertion-sensitive reasoning or category×condition interactions.

Clinical KG construction. Multi-LLM KG-RAG [19] uses multi-agent LLM prompting with schema-constrained extraction for oncology, modeling uncertainty via entropy but not tracking assertion classes on edges. AutoRD [20] applies GPT-4 to rare disease literature; RECAP-KG [21] handles messy GP notes; FHIR-Ontop-OMOP [22] generates virtual KGs from OMOP data. All share a common limitation: assertion status is not propagated into the final graph.

Assertion detection. NegEx [23] introduced trigger-based negation detection; ConText [24] extended it with temporality and experiencer. The i2b2/VA shared task [1] formalized a six-class taxonomy; Gul et al. [2] fine-tuned LLMs to 0.962 accuracy. All treat assertion detection as a terminal annotation task: labels are not carried into knowledge graphs or retrieval systems.

Assertion detection is necessary but not sufficient: labels must also be *temporally situated*—the same condition may be present in one encounter and absent in the next.

Temporal knowledge graphs. MedTKG [11] constructs temporal KGs with time-stamped snapshots (event time only); Graphiti [12] implements bitemporal edges [25] but lacks clinical ontology alignment; TEO [26] and TCL [27] formalize Allen's relations [28] as annotation ontologies or modal operators rather than KG edge attributes. Uncertainty quantification in biomedical KGs [29] has not been connected to assertion taxonomies. Two structural gaps emerge: an *epistemic propagation gap* (assertion-detecting systems do not persist labels into KGs) and a *temporal integration gap* (temporal formalisms operate as annotation layers, not as KG edge attributes participating in retrieval). EpiKG closes both by carrying assertion and temporal metadata as first-class properties through every pipeline stage.

3 System Design

3.1 Method Overview

EpiKG implements three key ideas: (1) end-to-end epistemic preservation, carrying assertion labels from NLP extraction through OMOP mapping, fact construction, KG materialization, and retrieval; (2) tri-temporal edge representation, storing valid time, transaction time, and NLP-asserted temporality on every KG edge; and (3) intent-aware routing, matching graph traversal strategy to question type. The first two ideas are necessary infrastructure; the third is where the performance gain originates. The key invariant is that *epistemic assertion status* is a first-class attribute at every stage, not an NLP annotation discarded after extraction (Figure 1). Clinical concepts are stored once as shared global nodes; patient-specific semantics (assertion status, temporality, confidence) reside on edges.

3.2 Epistemic Assertion Schema

Clinical notes abound with qualified statements—"*no evidence of pneumonia*," "*possible CHF*," "*mother had breast cancer*"—yet standard representations discard these qualifiers. OMOP excludes negated conditions entirely [3]; FHIR limits assertion metadata to conditions. EpiKG defines a seven-value assertion taxonomy:

$$\alpha \in \{\text{Pres., Abs., Poss., Cond., Hypo., Fam.Hx., Hist.}\} \tag{1}$$

This extends the i2b2 six-class taxonomy [1] by separating HISTORICAL from FAMILY_HISTORY, a clinically important distinction that prior work conflates (definitions in Appendix R). A rule-based classifier (122 scope-aware trigger patterns) assigns α with an associated confidence score. The label is carried through every stage: from `ExtractedMention` through ensemble merging, `ClinicalFact` records, KG edge properties, and retrieval scoring. Additionally, each edge carries

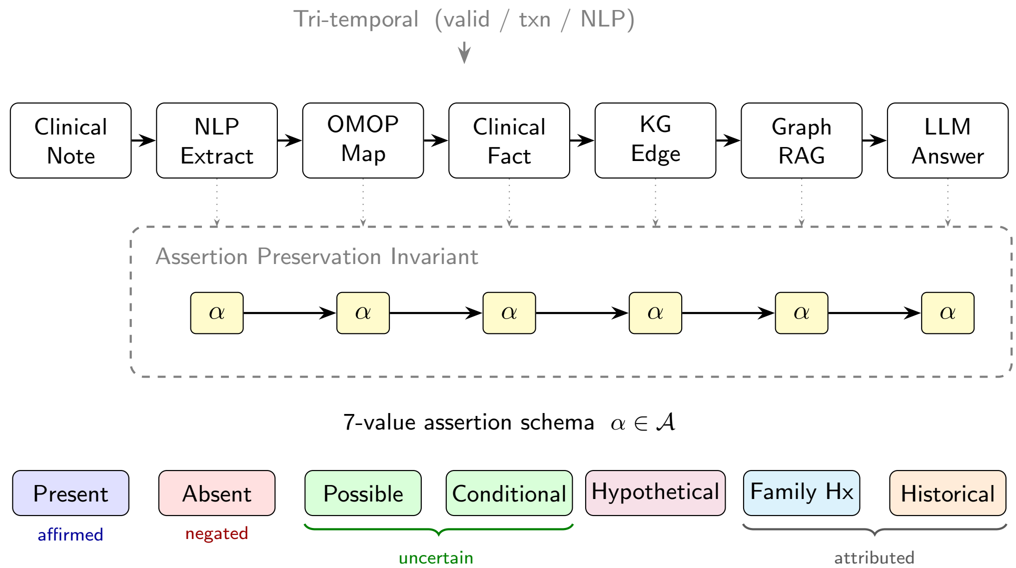

Figure 1: The EpiKG pipeline. The assertion label α (highlighted) is preserved as a first-class attribute from NLP extraction through OMOP mapping, fact construction, KG materialization, and assertion-aware GraphRAG.

a tri-temporal representation—valid time, transaction time, and NLP-asserted temporality—with Allen-style interval relations as first-class retrieval features (formalized in Appendix C.1).

3.3 Formal Epistemic Preservation

We formalize the epistemic invariant maintained by the EpiKG pipeline, building on the ontological principle that aboutness must be preserved across transformations [30]. The *epistemic state* of a clinical mention m is the tuple $e(m) = (c, \alpha, \xi, \tau)$: OMOP concept, 7-value assertion label, experiencer ($\{\text{PATIENT}, \text{FAMILY}\}$), and temporality ($\{\text{CURRENT}, \text{PAST}, \text{FUTURE}\}$); see Table 8 (Appendix) for notation summary. A pipeline *epistemically preserves* m if the output epistemic state equals the extraction state at every stage. We formalize (Appendix B) that an assertion-blind pipeline collapses the assertion distribution to a degenerate point mass, incurring entropy loss $\Delta H = H(A_c^\pi) \geq 0$ [31], and that assertion-faithful accuracy is bounded by $1 - f_{np}(c)$, where f_{np} is the fraction of non-present assertions. The empirical consequences are measured via category-stratified accuracy in Section 5.

3.4 Intent-Aware Retrieval (C4g)

The base retrieval pipeline uses bidirectional BFS over patient KG edges and OMOP vocabulary relationships (details in Appendix C.2), but treats all questions uniformly. Yet different clinical question types require fundamentally different graph operations. A *change* question requires cross-admission set differencing; a *current-state* question needs filtering to the most recent valid edges; a *historical* question must recover resolved conditions.

Intent classifier. A lightweight rule-based classifier maps each question to one of four intents—CHANGE, CURRENT_STATE, HISTORICAL, or DEFAULT—using question metadata with keyword-pattern fallback (Algorithm 1, Appendix T).

Change retrieval. For $\iota = \text{CHANGE}$, edges are partitioned by admission (`hadm_id`). The module computes set differences across admission pairs (k, k')—concepts *added* ($\mathcal{C}_{k'} \setminus \mathcal{C}_k$), *removed* ($\mathcal{C}_k \setminus \mathcal{C}_{k'}$), and *continued* ($\mathcal{C}_k \cap \mathcal{C}_{k'}$)—and returns structured evidence.

Current-state and historical retrieval. For CURRENT_STATE, the module filters to edges with $\tau_a = \text{CURRENT}$ or open-ended validity, deduplicates by concept, and emits "NOT FOUND" for

Table 1: ClinicalBench conditions. C1–C4g form the ablation ladder; C6 and C7 are bookend baselines.

ID	Condition	Retrieval	Assertion	Temporal
C1	LLM Alone	None	None	None
C2	+ Vanilla RAG (TF-IDF)	TF-IDF doc chunks	None	None
C2b	+ Vanilla RAG (dense)	Contriever doc chunks	None	None
C3	+ KG-RAG (no assertions)	Graph + Doc	None	None
C4	+ Epistemic KG-RAG	Graph + Doc	Full (7-class)	Tri-temporal
C4g	+ Intent-Aware KG-RAG	Graph + Doc (type-specific)	Full (7-class)	Tri-temporal
C6	Long Context	All notes	None	None
C7	Deterministic KG	KG lookup (no LLM)	Full	Tri-temporal

Table 2: ClinicalBench cohort demographics ($n = 43$ MIMIC-IV patients).

Characteristic	Category	n (%)
Sex	Female	26 (60.5%)
	Male	17 (39.5%)
Age (anchor)	18–29	2 (4.7%)
	30–49	13 (30.2%)
	50–64	13 (30.2%)
	65–79	9 (20.9%)
	80+	6 (14.0%)
Race/Ethnicity	White	33 (76.7%)
	Black/African American	6 (14.0%)
	Asian	1 (2.3%)
	Other/Unknown	3 (7.0%)
Insurance	Medicare	19 (44.2%)
	Private	15 (34.9%)
	Medicaid/Other	9 (20.9%)

missing concepts. For HISTORICAL, it selects edges with $\tau_a = \text{PAST}$ and augments with admission-based inference (concepts in earlier admissions but absent from the latest are labeled "resolved"). Each intent triggers a type-specific prompt template that structures evidence according to question semantics (Appendix V).

4 Benchmark Design

Existing clinical QA benchmarks evaluate factual recall (MedQA [32]) or agent task completion [33] but do not test handling of epistemic qualifiers or reasoning over longitudinal records of varying complexity. We introduce two complementary benchmarks built on MIMIC-IV [34].

4.1 ClinicalBench: Assertion-Sensitive Clinical QA

ClinicalBench comprises 400 gold-standard questions over 43 MIMIC-IV patients in two tasks: **Task A** (200 questions, negation-aware retrieval) and **Task B** (200 questions, temporal reasoning). Questions span 9 categories: *negation*, *conditional*, *uncertainty*, *family_history*, *sequence*, *current_state*, *duration*, *historical*, and *change*. Gold-standard answers were created by a single physician (board-certified emergency medicine) through manual chart review against source MIMIC-IV documents. No inter-annotator agreement was computed for gold-standard creation; the physician adjudication pilot (Section 4.4) addresses this limitation. Four ablation conditions progressively add system components (Table 1). Two bookend baselines—C6 (all notes concatenated) and C7 (deterministic KG lookup, no LLM)—bracket the design space.

Cohort demographics. The 43 ClinicalBench patients (Table 2) are drawn from a single academic medical center (Beth Israel Deaconess, Boston). The cohort skews female (60.5%), White (76.7%), and Medicare-insured (44.2%), reflecting the source institution's patient population. Subgroup accuracy analyses are underpowered given the small cohort; we report them transparently in Appendix Y rather than drawing subgroup conclusions.

Endpoints. The **pre-registered primary endpoint** is the hard longitudinal subset (*change* + *current_state* + *historical*, $n = 130$): C4g vs. C1 aggregate accuracy. The **secondary endpoint** is full 400-question ClinicalBench, C4g vs. C1. **Diagnostic endpoints** are per-category C1→C4g accuracy deltas and the C3→C4→C4g decomposition isolating assertion metadata (C3→C4) from intent routing (C4→C4g).

What each transition isolates. C1→C2/C2b isolates the *document retrieval effect*: C2 adds TF-IDF and C2b adds Contriever dense retrieval over clinical notes, analogous to existing medical RAG systems (DoctorRAG, MedRAG). C2/C2b→C3 isolates the *graph structure effect*: both retrieve documents, but C3 replaces document-level retrieval with graph traversal—testing whether structured retrieval outperforms vanilla retrieval. C3→C4 isolates the *assertion metadata effect*: C4 adds 7-class assertions and tri-temporal metadata but uses uniform BFS (no intent routing). C4→C4g isolates the *intent routing effect*: C4g adds question-type-specific graph traversal and assertion-aware filtering on top of C4's metadata. C4g vs. C6 isolates the *structured-vs-unstructured retrieval effect* on the same questions.

4.2 SliceBench: Complexity-Stratified Longitudinal QA

SliceBench tests the hypothesis that KG-augmented retrieval provides increasing benefit as patient record complexity grows. We select 6 MIMIC-IV patients in three tiers: **Tier A** (2 patients, 1–2 encounters), **Tier B** (2 patients, 5–10), and **Tier C** (2 patients, 15+). Each patient receives 24 questions (including hard longitudinal categories: cross-encounter medication timelines, problem list reconciliation, causal chain tracing), yielding 144 total. Five conditions (B0–B4) form a monotone context progression from parametric-only to full system (Appendix K). The critical comparison is B2→B3: both have the same documents, but B3 adds structured KG context with assertion metadata and temporal relations.

4.3 Evaluation Protocol

ClinicalBench uses a *deterministic keyword evaluator* (v2) that performs exact word-boundary matching of gold-standard assertion and temporal keywords against model answers (keyword lists in Appendix Q). The v2 evaluator includes an *abstention detection gate*: when a model responds that information is unavailable in the notes (e.g., "cannot determine," "not mentioned") rather than making a clinical claim, the answer is scored as incorrect regardless of keyword matches. This prevents false positives where refusal language incidentally matches gold-standard patterns. Sequence and change categories additionally require domain-specific keywords (ordering terms and change terms, respectively). This evaluator is fully reproducible (no stochastic LLM judge) and is used for all ClinicalBench results. The primary answering model is Claude Opus 4.6; cross-model validation uses MedGemma 27B and GPT-OSS 20B under the same evaluator.

SliceBench uses an *LLM-as-judge* protocol with separated answering and judging models to prevent self-evaluation bias [35]: Claude Sonnet 4.5 generates answers and Claude Opus 4.6 judges correctness.

Statistical reporting. We report BCa bootstrap 95% CIs ($n = 2{,}000$, seed 42) [36] on both benchmarks. Question-level resampling is the primary analysis; patient-level cluster bootstrap (resampling the 43 patients with replacement, including all their questions) is reported as a sensitivity analysis to account for within-patient correlation (Appendix Table 17). McNemar's test [37] provides a complementary non-parametric paired comparison; Benjamini–Hochberg FDR correction is applied to the family of McNemar tests (all remain significant after correction; the non-significant C3→C4 comparison, $p_{\mathrm{BH}} = 0.37$, is unaffected). ClinicalBench CIs appear on primary deltas (overall C4g−C1 and hard longitudinal subset); SliceBench CIs appear on all conditions (Table 6). A safety score with asymmetric weighting ($w = 2.0$ for false-positive assertion errors) captures clinical risk (Appendix E).

4.4 Physician Validation Protocol

To validate automated scoring, a board-certified physician (A.S.) reviewed a blinded, stratified pilot sample ($n = 30$ ClinicalBench items across all 9 categories and both conditions, C1 vs. C4g). The physician-rated C4g–C1 delta (+52 pp) exceeded the auto-evaluator delta (+47 pp), indicating the headline effect is conservative (Section 5.5). A second independent reviewer is planned; protocol details are in Appendix P.

Table 3: ClinicalBench primary results (400 questions, Claude Opus 4.6, keyword evaluator v2 with abstention detection). Ablation ladder (C1–C4g) progressively adds components; C4 isolates assertion metadata from intent routing; C6 and C7 are bookend baselines. Sig: * denotes BCa CI excluding zero. Best in **bold**.

	Condition	Accuracy	Δ vs C1
C1	LLM Alone	21.8%	—
C2	+ Vanilla RAG (TF-IDF)	52.2%	+30.5 pp *
C2b	+ Vanilla RAG (dense)	50.7%	+29.0 pp *
C3	+ KG-RAG (no assertions)	50.0%	+28.2 pp *
C4	+ Assertions (no routing)	46.8%	+25.0 pp *
C4g	+ Intent-Aware KG-RAG	**69.0%**	+47.2 pp *
C6	Long Context (all notes)	39.0%	+17.2 pp
C7	Deterministic KG (no LLM)	3.8%	−18.0 pp

5 Results

We evaluate EpiKG through two benchmarks: ClinicalBench tests assertion-sensitive and longitudinal reasoning, while SliceBench measures longitudinal QA across patient complexity tiers. ClinicalBench uses a deterministic keyword evaluator for reproducibility; SliceBench uses LLM-as-judge evaluation (Section 4.3). Claude Opus 4.6 is the primary answering model; MedGemma 27B, GPT-OSS 20B[1], and Qwen3.5 35B[2] provide cross-model validation. Both benchmarks report bias-corrected and accelerated (BCa) bootstrap 95% CIs ($n = 2{,}000$, seed 42) [36], resampled at the question level.

5.1 ClinicalBench: Primary Ablation

Intent-aware KG-RAG (C4g) reaches 69.0% (+47.2 pp vs. C1; 95% BCa CI: [+40.9 pp, +52.8 pp]; patient-level cluster bootstrap: [+40.2 pp, +52.5 pp]; McNemar $\chi^2 = 157.1$, $p < 10^{-6}$). Without retrieval, the model mostly abstains or produces vague responses that fail keyword matching; with structured retrieval, it produces substantive correct answers. The bookend baselines bracket the design space. C6 (all notes concatenated) scores 39.0%, outperforming C1 (+17.2 pp) because it provides clinical text. However, C6 fails on historical (0%) and sequence (0%), consistent with the "lost in the middle" phenomenon when context exceeds what the model can attend to. C7 (deterministic KG, no LLM) achieves 3.8%—only negation detection works (13.6%) via stored assertion metadata—confirming that an LLM is essential for reasoning over graph evidence.

Ablation decomposition: retrieval vs. assertions vs. intent routing. C2 (TF-IDF document retrieval) and C2b (dense retrieval via Contriever) score comparably—52.2% and 50.7%, respectively ($\Delta = -1.5$ pp; 95% BCa CI: [-6.6 pp, +3.5 pp]; McNemar $p = 0.62$)—confirming that the retrieval method (keyword vs. dense) does not explain the C2→C4g gap. C3 (graph-augmented retrieval without assertions) also scores similarly (50.0%), demonstrating that graph structure alone does not improve aggregate accuracy over either retrieval baseline. However, per-category patterns diverge: C3 excels on current state (54% vs. 32%) where graph structure helps identify active conditions, while C2 outperforms C3 on change (43% vs. 17%) where raw text explicitly mentions medication changes.

The new C4 condition isolates assertion metadata from intent routing: C4 adds epistemic annotations (negation, uncertainty, temporality) to graph edges but uses uniform BFS traversal without question-type-specific routing. C4 scores 46.8%—*below* C3 (50.0%)—with the C3→C4 delta of -3.2 pp not reaching significance (95% BCa CI: [-10.0 pp, +3.2 pp]; McNemar $p = 0.37$). This reveals that assertion metadata without intent routing does not improve—and may degrade—performance: the additional edge properties increase evidence verbosity without providing a mechanism to exploit them. The entire C3→C4g gain (+19.0 pp; CI: [+13.8 pp, +24.0 pp]) is attributable to intent-aware routing (C4→C4g: +22.2 pp; CI: [+16.0 pp, +28.0 pp]; McNemar $\chi^2 = 41.4$,

[1] GPT-OSS 20B is an open-source general-purpose model run locally via Ollama (4-bit quantized). We use the label GPT-OSS to avoid implying endorsement; the reproducibility package includes model identifiers and raw predictions.

[2] Qwen3.5 35B is an open-source reasoning model run locally via Ollama (4-bit quantized, thinking suppressed via template modification to ensure direct answers consistent with other models).

Table 4: ClinicalBench per-category accuracy (%) across conditions (Claude Opus 4.6, evaluator v2). C2b replaces TF-IDF with Contriever dense retrieval; C4 adds assertion metadata to C3 without intent routing; C4g adds intent-aware routing. Best per row in **bold**.

Category	n	C1	C2	C2b	C3	C4	C4g	C4g−C1
Negation	110	45.5	70.9	75.5	65.5	**90.0**	80.9	↑ 35.5 pp
Conditional	20	0.0	35.0	50.0	30.0	**50.0**	45.0	↑ 45.0 pp
Uncertainty	40	12.5	35.0	32.5	37.5	**52.5**	50.0	↑ 37.5 pp
Family hist.	30	3.3	43.3	43.3	43.3	50.0	**56.7**	↑ 53.3 pp
Sequence	40	10.0	60.0	47.5	57.5	10.0	**65.0**	↑ 55.0 pp
Current st.	50	26.0	32.0	32.0	54.0	30.0	**70.0**	↑ 44.0 pp
Duration	30	30.0	**66.7**	**66.7**	60.0	46.7	**66.7**	↑ 36.7 pp
Historical	50	6.0	48.0	46.0	42.0	12.0	**60.0**	↑ 54.0 pp
Change	30	6.7	43.3	20.0	16.7	10.0	**100.0**	↑ 93.3 pp
Overall	**400**	21.8	52.2	50.7	50.0	46.8	**69.0**	↑ 47.2 pp

$p < 10^{-10}$).

Per-category decomposition clarifies the mechanism: assertions without routing help on assertion-sensitive categories (negation: $+24.5$ pp, uncertainty: $+15.0$ pp) but degrade temporal categories (current state: -24.0 pp, historical: -30.0 pp, sequence: -47.5 pp). Intent routing reverses these losses and amplifies the gains—most dramatically on change ($+90.0$ pp), sequence ($+55.0$ pp), and historical ($+48.0$ pp). The finding that assertions do not help without routing—and that their value is unlocked specifically by intent-aware retrieval—validates the two-component architecture.

C2/C2b serve as the closest analogs to existing medical RAG systems (DoctorRAG, MedRAG), which use document retrieval without assertion-aware graph structure; the $+18.2$ pp C2b→C4g gap (CI: [+12.3 pp, +23.5 pp]; McNemar $p < 10^{-9}$) quantifies the value of epistemic KG-RAG over modern dense retrieval.

Evaluator hierarchy. We report three evaluators to bound the true effect size. The *primary* evaluator is the deterministic keyword matcher (v2 with abstention detection), chosen for reproducibility: it is fully deterministic and produces the same score on every run. Two *sensitivity analyses* probe its biases. First, a blinded physician pilot (Section 5.5) found the keyword evaluator too strict in 43% of cases (vs. too lenient in 10%); the physician-rated C4g–C1 delta ($+52$ pp) exceeds the keyword delta ($+47$ pp), indicating the headline effect is conservative. Second, an LLM-as-judge evaluation (Appendix W) achieves higher physician concordance (56.7% vs. 46.7%) and yields a C4g–C1 delta of $+28.5$ pp—lower because the judge credits partial answers and appropriate C1 abstentions that the keyword evaluator penalizes. The three deltas ($+28.5$ pp, $+47.2$ pp, $+52$ pp) bound the effect: the keyword evaluator understates C1 (conservative bias), while the physician-rated delta may overstate it given small n. All three agree that C4g substantially outperforms C1; the disagreement concerns magnitude, not direction.

5.2 Category×Condition Interaction

Table 4 and Figure 2 reveal a strong *category×condition interaction*: intent-aware KG-RAG improves all 9 categories over C1, with every delta exceeding $+35$ pp. Per-category results are diagnostic and exploratory; no multiplicity correction is applied. The preregistered primary endpoint is the hard longitudinal subset (change + current state + historical). Under the v2 evaluator with abstention detection, C1 drops sharply because the model without retrieval context frequently abstains ("insufficient information in the notes") rather than attempting an answer.

Where intent-aware routing excels. The largest gains occur on categories requiring cross-admission synthesis: change ($+93$ pp), sequence ($+55$ pp), historical ($+54$ pp), family history ($+53$ pp), conditional ($+45$ pp), and current state ($+44$ pp). On the hard longitudinal subset (change + current state + historical, $n = 130$), C4g outperforms C1 by $+59.2$ pp (95% BCa CI: [+49.2 pp, +67.7 pp]; patient-level: [+48.6 pp, +69.4 pp]). All nine category improvements exceed $+35$ pp (Appendix Figure 4, right panel); the left and middle panels decompose the C3→C4g gain into assertion metadata (C3→C4) and intent routing (C4→C4g), showing that assertions alone help assertion-

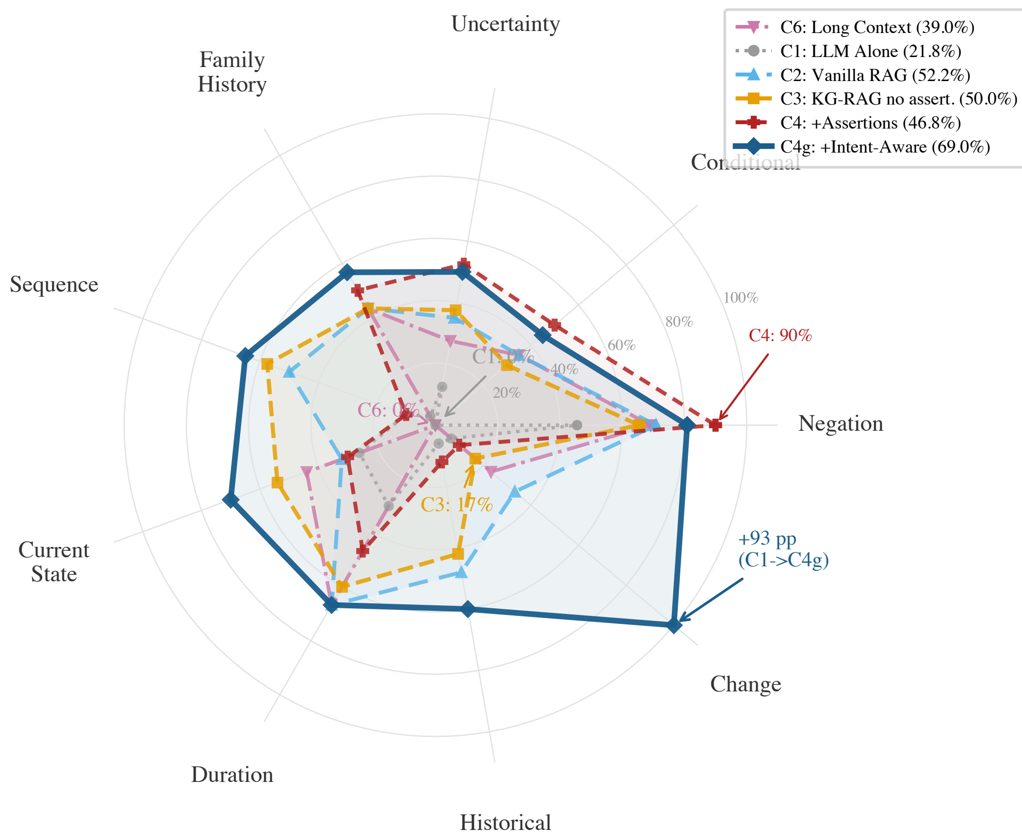

Figure 2: ClinicalBench per-category accuracy across six conditions (C6, C1, C2, C3, C4, C4g; Claude Opus 4.6, evaluator v2). C6 (pink dash-dot) fails on historical and sequence (both 0%); C1 (gray dotted) falls below 50% on 7 of 9 categories; C2 (light blue) and C3 (orange dashed) score similarly overall (∼50–52%) but diverge per category; C4 (firebrick, short-dash) adds assertions without routing and scores 46.8%—below C3—demonstrating that assertions alone are insufficient; C4g (dark solid) adds intent routing, unlocking the value of epistemic annotations.

Table 5: ClinicalBench cross-model results (400 questions, keyword evaluator v2). All four models benefit from KG-RAG; deltas vary by model. McNemar's test significant for all ($p < 10^{-6}$).

	Model	C1	C4g	Δ
Commercial	Claude Opus 4.6	21.8%	**69.0%**	+47.2 pp
Open	GPT-OSS 20B	20.5%	58.0%	+37.5 pp
Medical	MedGemma 27B	26.2%	57.8%	+31.5 pp
Reasoning	Qwen3.5 35B	37.0%	57.5%	+20.5 pp

sensitive categories but do not improve—and may degrade—temporal ones. C6 fails catastrophically on historical (0%) and sequence (0%) despite Opus's 200K window, confirming that long context cannot substitute for structured retrieval on these categories.

5.3 Cross-Model Validation

All four models benefit substantially from KG-RAG (Table 5), with C4g accuracies clustering between 57.5% and 69.0% despite C1 baselines ranging from 20.5% to 37.0%. Qwen3.5—a reasoning-optimized model—achieves the highest C1 baseline (37.0%) yet gains least (+20.5 pp; 95% BCa CI: [+14.9 pp, +25.8 pp]; McNemar $\chi^2 = 45.6$, $p < 10^{-11}$), while the two general-purpose models with lowest baselines (Opus 21.8%, GPT-OSS 20.5%) gain most (+47 pp and +38 pp). Across all four models, weaker baselines gain more from KG-RAG—a descriptive pattern consistent with a knowledge-compensator effect in which structured epistemic context partially substitutes for para-

Table 6: SliceBench overall results (144 questions, Claude Sonnet 4.5, judged by Opus 4.6). Sig: * denotes CI excluding zero.

	Condition	Score	95% CI	Δ vs B0
B0	LLM Alone	49.9%	[43.6%, 56.1%]	—
B1	Latest Note	73.4%	[67.8%, 78.7%]	+23.5 pp
B2	All Notes RAG	84.7%	[80.2%, 88.8%]	+34.8 pp *
B3	KG-RAG	86.9%	[82.8%, 90.8%]	+37.0 pp *
B4	Full System	87.6%	[83.7%, 91.2%]	+37.8 pp *

metric clinical knowledge ($n = 4$ models; the trend is descriptive and requires additional data points to confirm).

All four models improve on all 9 categories under C4g (per-category cross-model breakdowns in Appendix Table 10). The C4g convergence—four models from different training lineages (commercial, open-source, medical, reasoning) cluster within 12 pp under KG-RAG despite a 17 pp spread at baseline—suggests the epistemic KG provides a knowledge floor that partially compensates for model-specific weaknesses.

5.4 SliceBench

The incremental KG layer (B2→B3) contributes +2.2 pp overall (CI: [-1.5 pp, +5.9 pp]), not reaching aggregate significance. However, the KG benefit is complexity-dependent: negligible for simple cases (Tier A, 1–2 notes: +0.6 pp) but meaningful for complex longitudinal patients (Tier C, 15+ encounters: +5.0 pp), suggesting that structured retrieval's value scales with clinical complexity as document volume overwhelms chunk-level retrieval (Appendix M).

5.5 Physician Adjudication

A board-certified physician (A.S.) reviewed a stratified pilot sample ($n = 30$ ClinicalBench items, all 9 categories) under blinded conditions (condition labels randomized as A/B; unblinding performed post-review for analysis). Each item was rated on four dimensions: gold-standard correctness, model-answer correctness, clinical safety, and clinical utility. Full rubric definitions and the scoring interface are described in Appendix P.

Physician-rated accuracy. After unblinding, KG-RAG answers (C4g, $n = 16$) were rated correct in 81% of items vs. 29% for LLM-alone (C1, $n = 14$; Table 7), a +52 pp physician-rated delta—larger than the +47 pp auto-evaluator delta and directionally consistent.

Safety and utility. C4g answers were rated safe in 88% of cases (vs. 50% for C1), helpful in 81% (vs. 29%), and misleading in 6% (vs. 21%). No answers in either condition were rated potentially harmful.

Auto-evaluator concordance. The keyword evaluator agreed with physician judgment in 47% of items. Disagreement was asymmetric: in 43% of cases the auto-evaluator was too strict (scored incorrect when the physician rated correct or partially correct), while in only 10% was it too lenient. This conservative bias is consistent with the abstention penalty noted in Section 5.1: C1 answers that correctly decline to answer without context are penalized by the keyword matcher.

Gold-standard quality. Physician review found 40% of auto-generated gold standards required revision (30% incorrect, 10% needing revision), with errors concentrated in change detection (0/4 correct) and negation (0/2 correct) categories. Gold-standard errors are constant across conditions and therefore do not affect the relative C1–C4g comparison, but they indicate that auto-generated clinical benchmarks require physician validation—a finding with implications beyond this study.

Limitations. The pilot is limited by small per-condition samples ($n = 14/16$) and a single reviewer; a second independent reviewer is planned. Despite these constraints, the physician-rated delta (+52 pp) is directionally and magnitude-consistent with the automated delta (+47 pp) across all four dimensions measured.

Table 7: Physician adjudication pilot ($n = 30$). Single reviewer, blinded during scoring; condition labels unblinded post-review for stratified analysis.

Metric	C1 ($n = 14$)	C4g ($n = 16$)
Physician-rated correct	29%	81%
Partially correct	21%	6%
Incorrect	50%	12%
Safe	50%	88%
Minor concern	50%	12%
Potentially harmful	0%	0%
Helpful	29%	81%
Neutral	29%	12%
Not useful	21%	0%
Misleading	21%	6%

6 Discussion and Limitations

Why long context underperforms. Brute-force long context (C6) scores 39.0%, outperforming C1 (+17.2 pp) because it at least provides clinical text for the model to draw on, but it still falls far short of C4g (69.0%) and fails completely on historical (0%) and sequence (0%). C6 concatenates all notes chronologically without retrieval or selection; more sophisticated long-context strategies (hierarchical summarization, temporal segmentation) might perform better, but C6's category-level failures—consistent with the "lost in the middle" phenomenon [38]—demonstrate that raw context dumping cannot substitute for structured retrieval. Historical questions require distinguishing resolved from active conditions across admissions; sequence questions require strict temporal ordering—both are implicit in raw text but explicit in the epistemic KG. C6 handles change detection modestly (23%) because the relevant information appears explicitly in the text, but KG-RAG achieves 100% (+77 pp).

Model-dependent KG benefit. Across the four models tested, KG-RAG benefit shows a consistent inverse trend: models with lower C1 baselines gain more (Table 5). The two general-purpose models (Opus 21.8%, GPT-OSS 20.5%) gain +47 pp and +38 pp respectively; the medically fine-tuned model (MedGemma, 26.2%) gains +32 pp; and the reasoning-optimized model (Qwen3.5, 37.0%) gains least (+21 pp). Despite a 17 pp spread at baseline, all four models converge to 57.5–69.0% under C4g, consistent with the epistemic KG providing a knowledge floor that partially compensates for model-specific weaknesses. With four models from distinct training lineages, the convergence pattern is consistent across all observed data points, though $n = 4$ models is insufficient for statistical inference on the correlation; additional models are needed to confirm the trend. This has practical implications for RAG system design: the same retrieval pipeline may yield different gains depending on the downstream model, and single-model evaluation risks overstating or understating value.

When KG context matters. The complexity-dependent SliceBench effect (Tier C: +5.0 pp vs. Tier A: +0.6 pp) suggests structured epistemic context is most valuable when: (a) document volume overwhelms chunk-level retrieval, (b) questions require cross-encounter synthesis where the same concept appears with different assertion states, and (c) assertion status evolves across encounters.

The C4 ablation cleanly decomposes the C3→C4g gap (+19.0 pp) into two components: assertions alone (C3→C4: −3.2 pp, CI crossing zero) and intent routing (C4→C4g: +22.2 pp, $p < 10^{-10}$). This reveals a key architectural insight: assertion metadata does not improve aggregate accuracy without question-type-specific retrieval (CI crossing zero), and its value is unlocked only when paired with routing. Without routing, assertions increase evidence verbosity and do not improve temporal categories (historical: −30 pp, sequence: −47.5 pp, per-category effects exploratory); with routing, the same metadata enables near-perfect change detection (C4 10% → C4g 100%). The finding validates the two-component design and suggests that future work on assertion-aware clinical NLP should couple enriched metadata with structured retrieval strategies.

Uncertainty remains weakest. Uncertainty improves from 12.5% to 50.0% (+37.5 pp), but absolute accuracy remains low. Possible and suspected conditions are linguistically subtle ("consider statin," "may have CHF") and require pragmatic inference beyond keyword matching.

Implications beyond clinical NLP. The finding that metadata enrichment without retrieval alignment *hurts* performance may generalize beyond clinical KGs. Any domain where structured annotations—sentiment, stance, certainty, provenance—are extracted but not integrated into the retrieval policy faces an analogous propagation gap. Legal reasoning requires distinguishing holdings from dicta; financial analysis must separate confirmed earnings from guidance; scientific synthesis depends on whether a result was replicated or retracted. In each case, the design principle is the same: preserve domain-specific metadata end-to-end *and* route retrieval by query intent. The ClinicalBench ablation (C3→C4→C4g) provides a template for evaluating this principle in other domains.

Threats to validity and claim scope. Our claims are bounded by dataset, evaluator, and model scope. Data scope is limited: ClinicalBench uses 43 patients (400 questions) and SliceBench 6 patients (144 questions), all from MIMIC-IV [34]; tier-level SliceBench effects remain hypothesis-generating ($n = 2$ patients per tier), and B2→B3 crosses zero ([-1.5 pp, +5.9 pp]). Evaluator uncertainty is substantial: keyword scoring agrees with physician judgment in 47% of the pilot, LLM-as-judge reaches 57%, and the three evaluators produce different absolute deltas. We therefore report all three and focus conclusions on directional agreement, not a single "true" effect size. Benchmark label quality is imperfect: physician review found a 40% error/revision rate in auto-generated gold answers (concentrated in change and negation), and pilot adjudication used a single reviewer. Model dependence is real: gains vary by model (Table 5), so Opus-scale effects should not be assumed universal. This study is an ablation analysis rather than a head-to-head leaderboard against existing medical RAG systems [5, 6, 13]; C2 is used as a proxy baseline for document-level retrieval on our longitudinal task. Runtime variance from quantized local inference (e.g., MedGemma ± 10 pp across runs) and single-site provenance further limit external validity. Finally, this work does *not* establish clinical deployment readiness: it evaluates QA behavior under controlled benchmarks, not prospective patient outcomes.

Conclusion. The epistemic propagation gap is a real and measurable phenomenon: clinical NLP systems that extract assertion status but discard it before retrieval systematically misrepresent patient state. The key finding is that closing this gap requires two components: assertion metadata alone does not improve accuracy (C3→C4: -3.2 pp, CI crossing zero), but intent-aware routing unlocks its value (C4→C4g: $+22.2$ pp, $p < 10^{-10}$). This holds across four models from distinct training lineages, with a convergence pattern in which all four cluster at 57.5–69.0% under C4g despite divergent baselines—consistent with the epistemic KG providing a knowledge floor. The design principle—preserve epistemic metadata, route retrieval by question intent, evaluate by category$\times$condition interaction—is falsifiable, domain-agnostic, and immediately testable in any RAG system operating over annotated structured data.

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A Notation Summary

Symbol	Meaning
$\alpha \in \mathcal{A}$	Assertion label (7 values, Eq. 1)
$\boldsymbol{\tau}_v$	Valid time (event date, valid from/to)
$\boldsymbol{\tau}_t$	Transaction time (recorded at, doc date, created at)
τ_a	NLP-asserted temporality (Current / Past / Future)
$r \in \mathcal{R}$	Temporal relation (9 values, Table 20)
c	Confidence score $\in [0, 1]$
ξ	Experiencer (Patient / Family)
$e(m)$	Epistemic state tuple (c, α, ξ, τ)
q, π	Clinical question, patient
ι	Question intent (Change / Current_State / Historical / Default)
$f_{np}(c)$	Fraction of non-present assertions for concept c

Table 8: Key notation used throughout the paper.

B Formal Epistemic Preservation

We formalize the epistemic invariant maintained by the EpiKG pipeline, building on the principle that aboutness—the relationship between information artifacts and the entities they represent—must be preserved across transformations [30].

Definition 1 (Epistemic State)**.** *For a clinical mention m, the* epistemic state *is the tuple $e(m) = (c, \alpha, \xi, \tau)$, where c is the OMOP concept identifier, $\alpha \in \mathcal{A}$ is the 7-value assertion label (Eq. 1), $\xi \in \{\text{PATIENT}, \text{FAMILY}\}$ is the experiencer, and $\tau \in \{\text{CURRENT}, \text{PAST}, \text{FUTURE}\}$ is the temporality. A pipeline P* epistemically preserves *mention m if $P(e(m)) = e(m)$; that is, the epistemic state at the output of every pipeline stage is identical to the state at extraction.*

Proposition 2 (Assertion Entropy Loss)**.** *For concept c in patient π's record, let A_c^π denote the assertion distribution with empirical frequencies $p(\alpha_i) = n_i / \sum_j n_j$ over the $|\mathcal{A}|$ assertion classes. The assertion entropy is*

$$H(A_c^\pi) = -\sum_i p(\alpha_i) \log p(\alpha_i)\ . \tag{2}$$

An assertion-blind pipeline collapses all mentions to $\alpha = \text{PRESENT}$, yielding a degenerate distribution with $H = 0$. The information loss is $\Delta H = H(A_c^\pi) \geq 0$ [31], with $\Delta H > 0$ strictly whenever any mention carries a non-present assertion.

Proof. Under the assertion-blind mapping $\phi : \alpha_i \mapsto \text{PRESENT}$ for all i, the output distribution assigns probability 1 to PRESENT and 0 to all other classes, so $H(\phi(A_c^\pi)) = 0$. By non-negativity of Shannon entropy, $\Delta H = H(A_c^\pi) - 0 = H(A_c^\pi) \geq 0$, with equality iff all mentions already carry $\alpha = \text{PRESENT}$. ☐

Corollary 3 (Faithfulness Bound)**.** *Let $f_{np}(c)$ denote the fraction of mentions of concept c carrying a non-present assertion ($\alpha \neq \text{PRESENT}$). Without assertion labels, the maximum assertion-faithful accuracy for any downstream task conditioned on c is bounded by $1 - f_{np}(c)$. In clinical records where negated and uncertain mentions are prevalent—e.g., "no pneumonia," "possible CHF"—this bound can be substantially below 1.*

Proof. Without assertion labels, any predictor must assign a single assertion class to all mentions of c. Choosing PRESENT (the majority class in clinical text) yields accuracy $1 - f_{np}(c)$; the $f_{np}(c)$ non-present mentions are necessarily misclassified. ☐

C Extended System Design

C.1 Tri-Temporal Knowledge Graph Model

Clinical events unfold across multiple time dimensions that prior systems model incompletely [11, 12]. Bi-temporal databases [25] distinguish valid time from transaction time; EpiKG adds a third dimension (NLP-asserted temporality) and stores all three on every KG edge:

Definition 4 (Tri-Temporal Edge)**.** *An edge $e = (s, p, o, \alpha, \boldsymbol{\tau}_v, \boldsymbol{\tau}_t, \tau_a, r, c)$ where:*

- *s, o are source and target nodes; p is the predicate (one of 24 edge types);*
- *$\alpha \in \mathcal{A}$ is the assertion label (Eq. 1);*
- *$\boldsymbol{\tau}_v = (event\_date, valid\_from, valid\_to)$ is the **valid time** interval—when the relationship held in the real world;*
- *$\boldsymbol{\tau}_t = (recorded\_at, doc\_date, created\_at)$ is the **transaction time**—when it was recorded;*
- *$\tau_a \in \{\textsc{Current}, \textsc{Past}, \textsc{Future}\}$ is the **NLP-asserted temporality**;*
- *$r \in \mathcal{R}$ is an Allen interval algebra relation [28]; $c \in [0, 1]$ is temporal confidence.*

$\mathcal{R}$ comprises seven of Allen's 13 canonical relations [28] (merging symmetric pairs like Before/Meets) plus CONCURRENT and UNKNOWN (nine total; full mapping in Table 20). Unlike TEO [26] and TCL [27], which use Allen's relations as annotation-ontology classes or modal operators, EpiKG stores r and c directly on materialized edges, enabling efficient temporal filtering during traversal.

C.2 Graph-Augmented Retrieval Details

Given a clinical question q and patient π, the base retrieval pipeline: (1) extracts concepts from q via NLP + OMOP enrichment; (2) traverses 2–3 hops via bidirectional breadth-first search (BFS) over patient KG edges and OMOP vocabulary relationships (20M+ edges) via PostgreSQL common table expressions (CTEs) (Appendix L), pruning edges with $c < 0.3$; (3) groups edges into four temporal views (event timeline, current state, historical, conflicts); (4) retrieves matching guideline sections; (5) scores edges by:

$$\text{score}(e) = c(e) + \mathbb{1}[\text{type}(e) \in Q_{\text{rel}}] \cdot 0.2 + \mathbb{1}[\tau_a(e) = \textsc{Current}] \cdot 0.1 \tag{3}$$

where the bonuses for question-relevant edge types (0.2) and current temporality (0.1) were set by manual tuning on a development set of 20 questions. The surviving subgraph is serialized into structured text preserving assertion labels (e.g., "ABSENT: pneumonia"); and (6) composes graph evidence, temporal context, guidelines, and source documents into a single prompt.

D Gap Analysis

Table 9: Capability comparison across representative systems. ✓ = supported, ○ = partial, × = not supported.

System	Patient KG	OMOP mapping	Assertion (7-class)	Tri-temporal	Assert.-aware RAG	Experiencer	Allen's algebra	Multi-hop (≥3)
DoctorRAG [13]	×	×	×	×	×	×	×	×
GFM-RAG [5]	○	×	×	×	×	×	×	✓
MedRAG [14]	○	×	×	×	×	×	×	○
KARE [6]	○	×	×	×	○	×	×	✓
Multi-LLM KG [19]	×	✓	○	×	×	×	×	×
MedTKG [11]	✓	×	×	○	×	×	×	✓
Graphiti [12]	×	×	×	○	×	×	×	✓
MediGRAF [16]	✓	×	×	×	×	×	×	✓
EpiKG (ours)	✓	✓	✓	✓	✓	✓	✓	○

Partial marks indicate limited coverage: GFM-RAG, MedRAG, and KARE build population-level or hierarchical KGs (not per-patient longitudinal graphs); KARE uses KG-augmented retrieval but without assertion awareness; Multi-LLM KG models uncertainty via entropy but not the full assertion taxonomy; MedTKG and Graphiti use temporal annotations but not tri-temporal models. MediGRAF is the closest patient-level competitor, constructing per-patient medical graphs from clinical notes, but it does not preserve assertion status or temporal relations as edge attributes. EpiKG's partial mark on multi-hop traversal (≥ 3 hops) reflects a deliberate architectural trade-off: PostgreSQL CTE-based traversal provides ACID compliance but degrades beyond 2 hops (Appendix L).

E Safety Score

The safety score $S = 1 - \frac{1}{N}\sum_{i=1}^{N} w_i \cdot \mathbb{1}[\hat{y}_i \neq y_i]$ applies $w_i = 2.0$ for false-positive assertion errors (reporting a negated or absent condition as present) and $w_i = 1.0$ otherwise, capturing asymmetric clinical risk where acting on a falsely affirmed condition is more dangerous than missing a present one. We treat $w = 2.0$ as a reasonable default; sensitivity to this choice is a direction for future work.

F Cross-Model Per-Category Results

Table 10: ClinicalBench per-category accuracy (%) by model under C1 and C4g (intent-aware KG-RAG, evaluator v2). All three models improve on all 9 categories.

		Claude Opus 4.6			MedGemma 27B			GPT-OSS 20B		
Category	n	C1	C4g	Δ	C1	C4g	Δ	C1	C4g	Δ
Negation	110	45.5	80.9	↑35.5 pp	57.3	76.4	↑19.1 pp	46.4	80.0	↑33.6 pp
Conditional	20	0.0	45.0	↑45.0 pp	15.0	35.0	↑20.0 pp	0.0	15.0	↑15.0 pp
Uncertainty	40	12.5	50.0	↑37.5 pp	7.5	32.5	↑25.0 pp	5.0	30.0	↑25.0 pp
Family hist.	30	3.3	56.7	↑53.3 pp	3.3	40.0	↑36.7 pp	3.3	40.0	↑36.7 pp
Sequence	40	10.0	65.0	↑55.0 pp	7.5	47.5	↑40.0 pp	12.5	42.5	↑30.0 pp
Current st.	50	26.0	70.0	↑44.0 pp	32.0	58.0	↑26.0 pp	36.0	56.0	↑20.0 pp
Duration	30	30.0	66.7	↑36.7 pp	53.3	70.0	↑16.7 pp	16.7	70.0	↑53.3 pp
Historical	50	6.0	60.0	↑54.0 pp	0.0	46.0	↑46.0 pp	0.0	60.0	↑60.0 pp
Change	30	6.7	100.0	↑93.3 pp	0.0	76.7	↑76.7 pp	0.0	70.0	↑70.0 pp
Overall	**400**	**21.8**	**69.0**	↑**47.2** pp	**26.2**	**57.8**	↑**31.5** pp	**20.5**	**58.0**	↑**37.5** pp

G MedGemma Full Per-Category Results

Category	n	**C7**	**C1**	**C4g**
Negation	110	13.6	57.3	76.4
Conditional	20	0.0	15.0	35.0
Uncertainty	40	0.0	7.5	32.5
Family History	30	0.0	3.3	40.0
Sequence	40	0.0	7.5	47.5
Current State	50	0.0	32.0	58.0
Duration	30	0.0	53.3	70.0
Historical	50	0.0	0.0	46.0
Change	30	0.0	0.0	76.7
Overall	**400**	**3.8**	**26.2**	**57.8**

Table 11: MedGemma 27B ClinicalBench per-category accuracy (%, keyword evaluator v2) for three conditions. MedGemma gains +32 pp overall under C4g, with all 9 categories improving. Strongest improvements on change (+77 pp) and historical (+46 pp).

H Experiencer Attribute Propagation Ablation

The `experiencer` attribute distinguishes patient conditions from family member conditions; without it, the graph conflates the two, causing family history misattribution.

The fix improved family history by +10.0 pp with zero regressions on guard categories (negation, conditional, duration, sequence all unchanged), confirming that the experiencer attribute is load-bearing for assertion-sensitive reasoning.

Table 12: Impact of experiencer attribute propagation on Opus C4 (400 questions, prior evaluator run). Categories with no change omitted.

Category	Pre-fix	Post-fix	Δ
Family history	63.3%	**73.3%**	+10.0 pp
Uncertainty	50.0%	**55.0%**	+5.0 pp
Historical	34.0%	38.0%	+4.0 pp
Current state	46.0%	48.0%	+2.0 pp
Overall C4	**68.2%**	**70.2%**	+2.0 pp

I SliceBench Figures

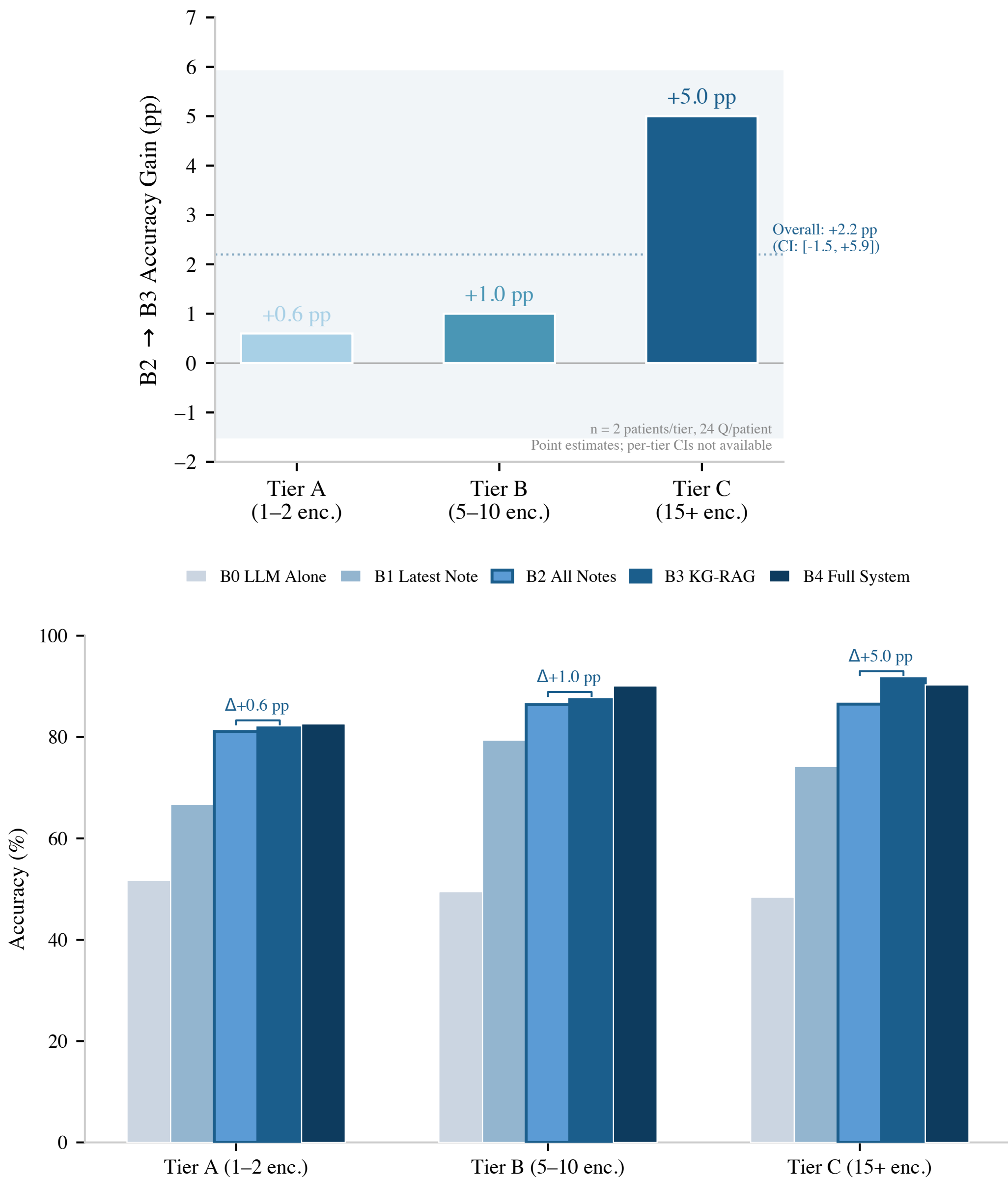

Figure 3: SliceBench results ($n = 2$ patients per tier, 24 questions each; point estimates without per-tier CIs). **Top:** The KG layer's incremental benefit (B2→B3) shows a directional, complexity-dependent trend: $+0.6$ pp (Tier A) to $+5.0$ pp (Tier C). **Bottom:** Full condition progression by tier. The overall B2→B3 delta is $+2.2$ pp (CI: [-1.5 pp, +5.9 pp]).

J Per-Category Delta Chart

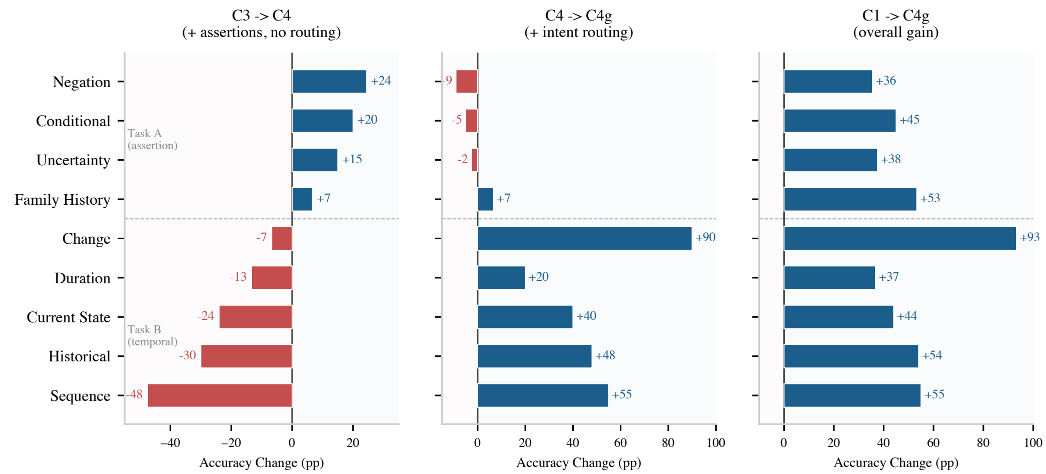

Figure 4: Per-category accuracy change (Claude Opus 4.6, evaluator v2) decomposing the C3→C4g gain. Categories sorted by C3→C4 delta; dashed line separates Task A (assertion-sensitive) from Task B (temporal). **Left:** C3→C4 (assertions without routing): assertion categories improve (negation +24 pp, conditional +20 pp) but temporal categories collapse (sequence −48 pp, historical −30 pp). **Middle:** C4→C4g (intent routing): routing recovers temporal categories (change +90 pp, sequence +55 pp, historical +48 pp) while slightly reducing assertion categories. **Right:** C1→C4g (overall): all 9 categories improve.

K SliceBench Conditions

Table 13: SliceBench conditions. B0–B4 form a monotone context progression.

ID	Condition	Context Provided to LLM
B0	LLM Alone	Question only (parametric knowledge)
B1	Latest Note	Most recent clinical note
B2	All Notes RAG	All notes, retrieved by relevance
B3	KG-RAG	Knowledge graph context + all notes
B4	Full System	KG-RAG + guidelines + calculators

L System Implementation Details

LLM baseline. On MedQA-USMLE (965 questions), Claude Opus 4.5 achieves 81.6% accuracy (5.1 pp below GPT-4 at 86.7% and Med-PaLM 2 at 86.5% [33]), establishing that the Claude model family—Sonnet 4.5 (SliceBench answerer) and Opus 4.6 (ClinicalBench primary, SliceBench judge)—provides a reasonable LLM baseline (Table 14).

Model	**Accuracy**	**Step 1**	**N**
EpiKG (LLM alone)	81.6%	79.9%	965
GPT-4 (2023)	86.7%	—	1,273
Med-PaLM 2 (2023)	86.5%	—	1,273
Claude 3 Opus (2024)	78.0%	—	1,273

Table 14: MedQA-USMLE results. Our LLM-alone baseline is competitive.

KG scale and traversal latency. The deployed system processes 145 documents across 85 patients, materializing 3,100 KG nodes and 8,803 edges. Graph traversal latency is sub-millisecond at this scale: 0.57 ms (1-hop), 0.75 ms (2-hop). The full system integrates 201 clinical calculators, 1,202 guideline sections, 20M+ OMOP vocabulary relationships, 122 assertion trigger patterns, and 24 edge types across 13 node types.

Multi-hop traversal. On the DR.KNOWS diagnostic reasoning benchmark [39], which measures KG traversal accuracy using PostgreSQL-backed clinical data, EpiKG achieves 0.420 overall (50%

at 1-hop, 25% at 2-hop, 0% at 3-hop). Multi-hop degradation reflects a deliberate architectural trade-off: PostgreSQL CTE-based traversal provides ACID compliance but scales poorly beyond 2 hops.

M SliceBench Tier-Stratified Results

Table 15: SliceBench results stratified by patient complexity tier. B2→B3 Δ shows the incremental KG contribution.

		Score (%)					
Tier	Encounters	B0	B1	B2	B3	B4	B2→B3 Δ
A	1–2 notes	51.7	66.7	81.1	81.8	82.6	$+0.6$ pp
B	5–10 notes	49.5	79.4	86.4	87.4	90.1	$+1.0$ pp
C	15+ notes	48.4	74.2	86.5	91.5	90.3	$+5.0$ pp

N ClinicalBench Full Per-Category Results (Opus)

Category	n	**C7**	**C6**	**C1**	**C2**	**C2b**	**C3**	**C4**	**C4g**
Negation	110	13.6	69.1	45.5	70.9	75.5	65.5	90.0	80.9
Conditional	20	0.0	35.0	0.0	35.0	50.0	30.0	50.0	45.0
Uncertainty	40	0.0	27.5	12.5	35.0	32.5	37.5	52.5	50.0
Family History	30	0.0	43.3	3.3	43.3	43.3	43.3	50.0	56.7
Sequence	40	0.0	0.0	10.0	60.0	47.5	57.5	10.0	65.0
Current State	50	0.0	44.0	26.0	32.0	32.0	54.0	30.0	70.0
Duration	30	0.0	66.7	30.0	66.7	66.7	60.0	46.7	66.7
Historical	50	0.0	0.0	6.0	48.0	46.0	42.0	12.0	60.0
Change	30	0.0	23.3	6.7	43.3	20.0	16.7	10.0	100.0
Overall	**400**	**3.8**	**39.0**	**21.8**	**52.2**	**50.7**	**50.0**	**46.8**	**69.0**

Table 16: Complete ClinicalBench per-category accuracy (%, Claude Opus 4.6, keyword evaluator v2) for all conditions. C2b (Contriever dense retrieval) scores comparably to C2 (TF-IDF), confirming the retrieval method does not drive the C2→C4g gap. C4 adds assertion metadata to C3 without intent routing; it helps assertion-sensitive categories (negation: $+24.5$ pp) but hurts temporal categories (historical: -30 pp, sequence: -47.5 pp). Only with intent routing (C4g) does the full system reach 69%.

O Physician Validation Pilot

A board-certified physician (A.S.) reviewed a stratified sample of $n = 30$ ClinicalBench items (14 C1, 16 C4g) spanning all 9 categories under blinded conditions. Each item was rated on: (1) gold-standard correctness (Yes / Partially / Needs Revision / No), (2) model-answer correctness (Correct / Partially Correct / Incorrect), (3) clinical safety (Safe / Minor Concern / Potentially Harmful), and (4) clinical utility (Helpful / Neutral / Not Useful / Misleading). Reviews were conducted using a custom scoring interface with access to the full de-identified MIMIC-IV discharge summaries, filtered to the admission(s) relevant to each question.

Gold-standard error analysis. Of the 30 gold standards reviewed, 18 (60%) were rated correct, 3 (10%) needed revision, and 9 (30%) were incorrect. Errors were concentrated in change detection (0/4 gold standards correct—NLP misread discharge medication lists) and negation (0/2 correct—NLP missed physical exam findings). These errors are constant across conditions and do not affect the relative C1–C4g comparison.

Auto-evaluator concordance. The keyword evaluator agreed with physician judgment in 47% of cases. In 43% of disagreements the evaluator was too strict (marked incorrect when the physician rated correct or partially correct); in 10% it was too lenient. This conservative bias largely reflects the abstention penalty: C1 answers that appropriately decline to answer without patient context are penalized by keyword matching.

P Physician Validation Protocol

Design. A board-certified physician independently scored ClinicalBench answers under two conditions (C1 and C4g) using a multi-dimensional rubric (correctness, safety, utility). The reviewer was blinded to condition assignment (conditions randomized as A/B). The pilot sample comprises 30 items stratified across all 9 categories and both conditions, selected to maximize coverage within a feasible review budget ($\sim$8 hours physician time).

Endpoints. (1) Human–keyword evaluator concordance rate (fraction of automated scores confirmed by physician). (2) Physician-rated C4g vs. C1 accuracy, safety, and utility. (3) Gold-standard error rate.

Disclosure. The reviewing physician (A.S.) is a co-author. Single-reviewer design limits inter-rater reliability assessment; a second reviewer is planned for the extended validation.

Status. This protocol was designed before examining primary results. Pilot results ($n = 30$) are reported in Section 5.5.

Q Reproducibility Details

Models. ClinicalBench primary ablation: Claude Opus 4.6 (claude-opus-4-20250514, keyword evaluator v2). Cross-model: MedGemma 27B (gemma3:27b, 4-bit GGUF) and GPT-OSS 20B (4-bit GGUF) via Ollama; same evaluator. SliceBench: Claude Sonnet 4.5 (claude-sonnet-4-5-20250929) answers, Opus 4.6 judges. Temperature 0 throughout; 4-bit quantization introduces GPU non-determinism (±10 pp run-to-run for MedGemma), controlled via within-run paired comparisons.

Evaluation. The deterministic keyword evaluator performs exact word-boundary matching (regex `\bKEYWORD\b`) against gold-standard assertion and temporal keywords. Per-category keyword sets: negation ("no," "denies," "absent," "negative"), uncertainty ("possible," "suspected," "may," "likely," "consider"), temporal ("before," "after," "during," "changed," "new"), plus category-specific terms. Evidence preambles (echoed graph context) are stripped before matching.

Retrieval. C2's vanilla RAG uses TF-IDF over chunked clinical notes (512-token chunks, 64-token overlap), retrieving the top-5 chunks by cosine similarity. C2b replaces TF-IDF with Contriever [40] dense embeddings (256-token chunks, 64-token overlap, top-k up to 6,000 characters); C2b scores 50.7% ($\Delta = -1.5$ pp vs. C2; $p = 0.62$), confirming the retrieval method does not explain the C2→C4g gap. These are the closest analogs to existing medical RAG systems (DoctorRAG, MedRAG) on patient-level longitudinal data. ClinicalBench conditions map to SliceBench: C1$\approx$B0, C2$\approx$B2, C4g$\approx$B3, C5$\approx$B4.

Bootstrap. $n = 2{,}000$ resamples, seed 42, BCa method [36]. We report both question-level and patient-level (cluster) bootstrap CIs; the latter resample the 43 patients with replacement, including all their questions per draw, to account for within-patient correlation. Patient-level CIs are slightly wider but all primary endpoints remain significant (Table 17).

McNemar's test. We supplement bootstrap CIs with McNemar's test for paired nominal data, comparing discordant pairs (C1 wrong/C4g right vs. C1 right/C4g wrong). All three pairwise comparisons are significant: C1 vs. C4g ($\chi^2 = 157.1$, $p < 10^{-6}$; discordant: 18 vs. 207), C1 vs. C3 ($\chi^2 = 73.4$, $p < 10^{-6}$; 29 vs. 142), and C3 vs. C4g ($\chi^2 = 49.3$, $p < 10^{-11}$; 19 vs. 95).

Data. De-identified MIMIC-IV clinical notes accessed under a PhysioNet Credentialed Health Data Use Agreement (v1.5.0).

Compute. ClinicalBench Opus C1/C3/C4g: $\sim$22min each (API); Opus C6: $\sim$3.5h (API); C7: $<$ 1min (no LLM); MedGemma C1–C4g: $\sim$3.6h (single GPU); MedGemma C6: $\sim$1.5h; SliceBench: $\sim$2h (API).

Evaluator evolution. The keyword evaluator underwent three versions: v0 (substring matching), v1 (word-boundary matching + evidence preamble stripping), and v2 (+ abstention detection gate + domain-specific keyword requirements for sequence and change). The v1 evaluator awarded false positives when models responded with "insufficient information in the notes" because negation and

Table 17: Statistical summary: question-level and patient-level (cluster) BCa bootstrap 95% CIs for primary endpoints. Both levels exclude zero for all comparisons.

Comparison	Δ	Question CI	Patient CI	McNemar p
C4g−C1 (overall)	+47.2 pp	[+40.9 pp, +52.8 pp]	[+40.2 pp, +52.5 pp]	$< 10^{-6}$
C2b−C2 (dense vs. keyword)	−1.5 pp	[-6.6 pp, +3.5 pp]	[-5.9 pp, +3.0 pp]	0.62
C4g−C2b (KG-RAG vs. dense)	+18.2 pp	[+12.3 pp, +23.5 pp]	[+12.9 pp, +23.2 pp]	$< 10^{-9}$
C3−C1 (retrieval)	+28.2 pp	[+22.3 pp, +34.0 pp]	[+23.1 pp, +33.8 pp]	$< 10^{-6}$
C4−C3 (assertions)	−3.2 pp	[-10.0 pp, +3.2 pp]	[-10.8 pp, +3.4 pp]	0.37
C4g−C4 (routing)	+22.2 pp	[+16.0 pp, +28.0 pp]	[+14.8 pp, +29.3 pp]	$< 10^{-10}$
C4g−C3 (both)	+19.0 pp	[+13.8 pp, +24.0 pp]	[+14.7 pp, +23.5 pp]	$< 10^{-11}$
Hard longitudinal	+59.2 pp	[+49.2 pp, +67.7 pp]	[+48.6 pp, +69.4 pp]	—

temporal keywords in the refusal text matched gold-standard patterns. The v2 evaluator adds an abstention detection layer: answers matching abstention patterns (e.g., "cannot determine," "not mentioned," "insufficient evidence") are scored as incorrect unless they contain clinical claim patterns (e.g., "patient does not," "denies"). Additionally, v2 requires sequence answers to contain ordering keywords ("first," "then," "before") and change answers to contain change keywords ("added," "removed," "discontinued"), preventing term-overlap-only false positives. The duration category's minimum-0.5 score floor for matching duration keywords was removed. The v1→v2 transition reduced C1 accuracy (from ∼50% to 21.8% for Opus) by correctly classifying abstention responses as incorrect; C4g accuracy decreased modestly (from ∼76% to 69.0%) because the model abstains less frequently when retrieval context is provided. All main-text numbers use v2.

Reproducibility package. A standalone reproducibility package is included as supplementary material (`epikg-benchmark/`). It contains all 400 ClinicalBench questions with gold answers, raw model predictions for all LLM-based conditions (Opus C1/C2/C3/C4/C4g/C6, MedGemma C1/C4g, GPT-OSS C1/C4g), scored outputs for the deterministic baseline (C7), and the keyword evaluator v2 with abstention detection. Running the evaluator requires only Python 3.10+ with no external dependencies. All accuracy numbers in Tables 3, 4, and 5 are independently reproducible from this package. MIMIC-IV patient identifiers are included so that PhysioNet-credentialed reviewers can trace predictions back to source clinical notes. The full ClinicalBench dataset—including all questions, predictions, and evaluation code—is publicly available on HuggingFace at `https://huggingface.co/datasets/alexstinard/epikg-clinicalbench` with Croissant metadata for machine-readable dataset discovery.

Results provenance. Table 18 maps each reported result to its source checkpoint file. MedGemma C4g has 399 predictions (1 timeout); the missing question is scored as incorrect ($n = 400$).

Table 18: Results provenance: checkpoint files for each condition×model combination. All scored with keyword evaluator v2.

Condition	Model	Checkpoint file	n
C1	Opus 4.6	`opus/C1_llm_alone.json`	400
C2	Opus 4.6	`opus/C2_vanilla_rag.json`	400
C2b	Opus 4.6	`opus/C2b_dense_rag.json`	400
C3	Opus 4.6	`opus/C3_kg_rag.json`	400
C4	Opus 4.6	`opus/C4_epistemic_kg_rag.json`	400
C4g	Opus 4.6	`opus/C4g_intent_aware.json`	400
C6	Opus 4.6	`opus/C6_long_context.json`	400
C7	—	`opus/C7_deterministic.json`	400
C1	MedGemma 27B	`medgemma/C1_llm_alone.json`	400
C4g	MedGemma 27B	`medgemma/C4g_intent_aware.json`	399*
C1	GPT-OSS 20B	`gptoss/C1_llm_alone.json`	400
C4g	GPT-OSS 20B	`gptoss/C4g_intent_aware.json`	400

*MedGemma C4g: 1 timeout; missing answer scored as incorrect for $n = 400$ denominator.

R Assertion Category Definitions

Assertion	Definition	Example
PRESENT	Condition currently affirmed	"has diabetes"
ABSENT	Condition explicitly negated	"denies chest pain"
POSSIBLE	Condition suspected, not confirmed	"possible pneumonia"
CONDITIONAL	Contingent on specific circumstances	"if febrile, start antibiotics"
HYPOTHETICAL	Discussed as a scenario	"would need dialysis if. . . "
FAMILY_HISTORY	Attributed to a family member	"mother had breast cancer"
HISTORICAL	Previously true, not necessarily current	"former smoker"

Table 19: The 7-value assertion taxonomy, extending i2b2 by separating HISTORICAL from FAMILY_HISTORY.

S Temporal Relation Mapping

The nine temporal relations $\mathcal{R}$ used on KG edges are derived from Allen's 13 canonical interval relations [28] by merging symmetric pairs.

$\mathcal{R}$ **value**	**Allen source(s)**	**Meaning**
BEFORE	Before, Meets	A ends before B starts
AFTER	After, Met-by	A starts after B ends
DURING	During	A entirely within B
CONTAINS	Contains	A entirely contains B
OVERLAPS	Overlaps, Overlapped-by	A and B partially overlap
STARTS	Starts, Started-by	A and B share start time
FINISHES	Finishes, Finished-by	A and B share end time
CONCURRENT	Equals	A and B have same interval
UNKNOWN	—	Relation undetermined

Table 20: Mapping from Allen's 13 interval relations to the 9 temporal relation values stored on KG edges.

T Intent-Aware Routing Algorithm

The C4g classifier is rule-based, mapping questions to one of four intents using keyword patterns (e.g., "changed", "new since" trigger CHANGE; "currently", "active problem" trigger CURRENT_STATE). Algorithm 1 details the full retrieval procedure.

Algorithm 1: Intent-Aware Retrieval (C4g)

Input: Question q, patient π
Output: Structured evidence E

$\mathcal{C} \leftarrow \textsc{ExtractConcepts}(q)$; `// NLP + OMOP enrichment`
$\iota \leftarrow \textsc{ClassifyIntent}(q)$; `//` $\in \{\texttt{Change}, \texttt{CurrSt}, \texttt{Hist}, \texttt{Default}\}$
if $\iota = \textsc{Change}$ **then**
  Partition edges by admission: $\mathcal{E}_k \leftarrow \{e \mid \texttt{hadm\_id}(e) = k\}$;
  foreach *admission pair* (k, k') *with* $k < k'$ **do**
    $\mathcal{A} \leftarrow \mathcal{C}_{k'} \setminus \mathcal{C}_k$; $\mathcal{R} \leftarrow \mathcal{C}_k \setminus \mathcal{C}_{k'}$; $\mathcal{S} \leftarrow \mathcal{C}_k \cap \mathcal{C}_{k'}$;
  $E_g \leftarrow \textsc{FormatChange}(\mathcal{A}, \mathcal{R}, \mathcal{S})$;
else if $\iota = \textsc{CurrSt}$ **then**
  $E_g \leftarrow \textsc{FilterEdges}(\pi, \mathcal{C},\ \tau_a = \textsc{Current} \vee \text{open validity})$;
  Deduplicate by concept; emit "NOT FOUND" for missing $c \in \mathcal{C}$;
else if $\iota = \textsc{Hist}$ **then**
  $E_g \leftarrow \textsc{FilterEdges}(\pi, \mathcal{C},\ \tau_a = \textsc{Past})$;
  Augment: concepts in earlier admissions but absent from latest $\rightarrow$ "resolved";
else
  $E_g \leftarrow \textsc{BidirectionalBFS}(\pi, \mathcal{C},\ \text{hops} = 2\text{–}3,\ c_{\min} = 0.3)$;
$E_d \leftarrow \textsc{RetrieveDocuments}(\pi, \mathcal{C})$;
return $\textsc{Compose}(E_g, E_d,\ \text{template}(\iota))$;

U Bookend and Diagnostic Conditions

C5 (Full System). C5 extends C4 with guideline retrieval (1,202 sections) and clinical calculators (201, e.g., CHA_2DS_2-VASc, MELD). In a prior evaluator run, C5 achieved 68.2% overall, below the prior-run C1 and C4, likely because non-relevant components dilute the context window.

C6 (Long Context). All patient documents are concatenated chronologically and presented to the LLM. For Opus (200K token window), all notes fit; for MedGemma (8K window), later documents are truncated. C6 achieves 39.0% (Opus) overall—better than C1 (21.8%) but far below C4g (69.0%). Opus scores 0% on historical and sequence under C6, confirming that brute-force long context cannot substitute for structured retrieval on categories requiring cross-encounter synthesis.

C7 (Deterministic KG). KG edges matching query concepts are returned directly without LLM reasoning. C7 achieves 3.8% overall (only negation at 13.6% via stored assertion metadata), confirming that structured data without an LLM reasoner cannot answer clinical questions.

V Illustrative Retrieval Example

To illustrate why intent-aware routing matters, consider a change question: *"What medications changed between this patient's first and second admissions?"*

C1 (LLM alone). The model sees only the current note, which mentions metoprolol, atorvastatin, and "discontinued lisinopril." Without prior admission records, it cannot determine what was *added* vs. *continued*, and may hallucinate prior medication lists.

C4g (intent-aware KG-RAG). The intent classifier routes to CHANGE retrieval, which partitions KG edges by `hadm_id` and computes set differences:

- **Added** (admission 2 only): atorvastatin 40mg [PRESENT]
- **Removed** (admission 1 only): lisinopril 10mg [PRESENT→ABSENT]
- **Continued**: metoprolol 25mg [PRESENT, both admissions]

The assertion labels (PRESENT, ABSENT) disambiguate: "discontinued lisinopril" is not a current medication but a historical one whose status changed—exactly the distinction the epistemic schema preserves.

W LLM-as-Judge Concordance

To complement the deterministic keyword evaluator, we rescored all 800 predictions (400 questions $\times$ 2 conditions: C1 and C4g) using Claude Opus 4.6 as an LLM judge. The judge prompt presents the question, reference answer, and system answer, and requests a score of 1 (correct), 0.5 (partially correct), or 0 (incorrect) with a one-sentence justification. Table 21 summarizes the concordance.

Table 21: Evaluator concordance comparison. LLM judge scores ≥ 0.5 treated as correct for binary comparison. Physician concordance computed on $n = 30$ overlapping items.

Metric	Keyword v2	LLM Judge
vs. Keyword evaluator ($n = 800$)		
Agreement	—	78.9%
Cohen's κ	—	0.572
vs. Physician ($n = 30$)		
Agreement	46.7%	56.7%
Too strict	43.3%	36.7%
Too lenient	10.0%	6.7%
Condition-level accuracy		
C1 accuracy	21.8%	28.5%
C4g accuracy	69.0%	57.0%
C4g$-$C1 Δ	+47.2 pp	+28.5 pp

Key findings. The LLM judge achieves higher physician concordance (56.7% vs. 46.7%) and is less prone to false strictness (36.7% vs. 43.3%), confirming that the keyword evaluator's conservative bias inflates the measured delta. Under the LLM judge, C1 rises from 21.8% to 28.5% (the judge credits clinically reasonable hedging that keyword matching penalizes), while C4g drops from 69.0% to 57.0% (the judge penalizes partial answers that pass keyword matching via term overlap). The resulting C4g$-$C1 delta under LLM judge (+28.5 pp) is lower than the keyword delta (+47.2 pp) but remains substantial and directionally consistent. The per-condition asymmetry—C1 gains, C4g loses—is consistent with the physician finding that the keyword evaluator disproportionately penalizes C1 abstentions.

Per-category analysis. The LLM judge is particularly stricter on change (C4g: 62% mean score vs. 100% keyword) because it penalizes partial medication lists that contain correct change keywords but miss specific drugs. Conversely, the judge is more generous on historical (C4g: 62% vs. 60% keyword) and negation (C4g: 79% vs. 81% keyword), where its semantic understanding better captures correct answers that use varied phrasing.

Limitations. Using the same model family (Opus) for both answering and judging introduces potential self-preference bias. The judge also shows moderate agreement with the keyword evaluator ($\kappa = 0.572$), indicating they capture partially overlapping but distinct aspects of answer quality.

X Assertion Classifier Evaluation

The rule-based assertion classifier uses 122 calibrated trigger patterns extending the NegEx [23] and ConText [24] frameworks to a 7-class taxonomy (Table 19). Table 22 summarizes the pattern inventory and confidence ranges by category.

Corpus statistics. Across the 43 ClinicalBench patients, the knowledge graph contains 3,943 edges linked to clinical facts. Of these, 618 (15.7%) carry non-present assertions: 442 absent (11.2%), 94 possible (2.4%), 71 historical (1.8%), 8 conditional (0.2%), and 3 hypothetical (0.1%). At the mention level, 1,428 of 12,379 mentions (11.5%) are non-present. This non-present fraction $f_{np} = 0.157$ provides the empirical bound from Corollary 3: an assertion-blind pipeline cannot exceed $1 - f_{np} = 84.3\%$ assertion-faithful accuracy on concepts where negated or uncertain mentions are present.

Table 22: Assertion trigger pattern inventory. Confidence ranges reflect per-trigger calibration.

Category	Patterns	Confidence	Example triggers
Absent	31	0.85–0.98	"no evidence of," "denies," "ruled out"
Uncertain	32	0.35–0.75	"possible," "likely," "consistent with"
Present	27	0.85–0.98	"confirmed," "diagnosed with," "positive for"
Hypothetical	11	0.25–0.35	"risk of," "screening for," "prophylaxis"
Family history	8	0.85–0.95	"family history of," "maternal history"
Conditional	7	0.20–0.40	"if," "contingent on," "depending on"
Historical	6	0.75–0.90	"history of," "former," "remote history"
Total	**122**	0.20–0.98	

Intrinsic evaluation. We sampled 189 mentions from the 43 ClinicalBench patients, stratified by predicted assertion type (50 present, 51 absent, 15 possible, 28 historical, 19 conditional, 4 hypothetical, 10 family history), and a physician (A.S.) annotated gold-standard labels in a blinded, randomized review. Table 23 reports per-class precision, recall, and F1.

Table 23: Intrinsic assertion classifier evaluation on 189 physician-annotated MIMIC-IV mentions (stratified sample from 43 ClinicalBench patients).

Assertion	n	P	R	F1
Present	62	0.980	0.790	0.875
Absent	51	0.980	0.961	0.970
Possible	15	0.500	1.000	0.667
Historical	28	0.933	1.000	0.966
Conditional	19	1.000	0.789	0.882
Hypothetical	4	1.000	1.000	1.000
Family history	10	0.900	0.900	0.900
Weighted avg	189	0.933	0.894	0.902

Overall accuracy is 89.4% (169/189; 95% Wilson CI: [84.2%, 93.0%]) with Cohen's $\kappa = 0.867$ (strong agreement). Negation detection achieves F1 = 0.970, consistent with published benchmarks: NegEx P = 94.5%/R = 77.8% [23], NegBio P = 96.3%/R = 85.7% [41]. The dominant error pattern (11/20 errors) is over-triggering uncertainty: the classifier assigns POSSIBLE to mentions near hedging language ("likely," "concerning for") that physicians judge as PRESENT. This explains the low precision for the POSSIBLE class (P = 0.50) despite perfect recall.

Functional evaluation. Rather than relying solely on intrinsic metrics, the C4 ablation provides a functional evaluation of assertion quality. C4 adds assertion metadata to C3's graph without intent routing: the classifier's output is directly consumed by the retrieval pipeline. On assertion-sensitive categories (negation, conditional, uncertainty, family history), C4 outperforms C3 by an average of $+16.5$ pp, confirming that the classifier produces usable assertions for these categories. On temporal categories (historical, sequence, change, current state, duration), C4 underperforms C3 by an average of -24 pp—not because assertions are incorrect, but because uniform BFS traversal cannot exploit them effectively. The C4→C4g routing correction recovers these losses and amplifies the gains, achieving $+22.2$ pp overall ($p < 10^{-10}$).

Y Cohort Demographics and Subgroup Analysis

The 43 ClinicalBench patients are drawn from MIMIC-IV [34], a single academic medical center (Beth Israel Deaconess Medical Center, Boston). Table 2 in the main text summarizes sex, age, race/ethnicity, and insurance distributions. The cohort skews female (60.5%), White (76.7%), and Medicare-insured (44.2%), reflecting the source institution's patient mix.

Subgroup accuracy. Table 24 reports C1 and C4g accuracy by demographic group. Because questions are distributed unevenly across patients and group sizes are small ($n \leq 26$ patients per stratum), these comparisons are severely underpowered; we report them for transparency, not for drawing subgroup conclusions. No statistically significant interaction between demographic group and condition was observed, but the analysis cannot rule out meaningful effect modification.

Table 24: ClinicalBench accuracy by demographic subgroup (underpowered; reported for transparency). n_q = number of questions in each stratum.

Characteristic	Group	n_{pts}	n_q	C1 (%)	C4g (%)
Sex	Female	26	249	22.5	69.9
	Male	17	151	20.5	67.5
Age	<65	28	258	20.5	69.4
	≥65	15	142	23.9	68.3
Race	White	33	290	22.4	71.7
	Non-White	10	110	20.0	61.8

Generalizability limitations. The single-site, predominantly White cohort limits external validity. MIMIC-IV emergency department notes are heavily templated; performance on narrative-heavy specialties (psychiatry, palliative care) or community hospital documentation styles is unknown. Multi-site validation with demographically diverse cohorts is needed before deployment claims can be made.

Z Ethics Statement

This work uses de-identified clinical data from MIMIC-IV [34] under a PhysioNet Credentialed Health Data Use Agreement. No patient re-identification was attempted. The system is designed for clinical decision *support*, not autonomous clinical decision-making.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: **Yes.**

   Justification: The abstract and introduction state four specific contributions (Section 1), each supported by empirical evidence in Sections 5.1–5.4. SliceBench claims include bootstrap CIs; ClinicalBench claims use within-run paired comparisons (Section 4.3). Claims are scoped with "to our knowledge" qualifiers where appropriate.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: **Yes.**

   Justification: Section 6 explicitly discusses limitations including small sample sizes (6 patients for SliceBench, 2 per tier), single-site data (MIMIC-IV), the B2→B3 confidence interval crossing zero, model-dependent effects, and physician pilot adjudication ($n = 30$, Section 5.5).

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (correct) proof?

   Answer: **Yes.**

   Justification: Proposition 2 (assertion entropy loss) and Corollary 3 (faithfulness bound) in Appendix B state all assumptions explicitly. These are information-theoretic characterizations with all steps included.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper?

   Answer: **Yes.**

   Justification: Appendix Q reports exact model versions (MedGemma 27B, Claude Sonnet 4.5, Claude Opus 4.6), temperature settings (0), bootstrap parameters ($n = 2{,}000$, seed 42, BCa method), and data source (MIMIC-IV v1.5.0 under PhysioNet DUA). Section 4 describes question generation and evaluation protocols.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?

   Answer: **Yes.**

   Justification: A standalone reproducibility package is included as supplementary material (`epikg-benchmark/`) containing all 400 questions, raw predictions for all condition$\times$model combinations (7 Opus conditions, 2 MedGemma, 2 GPT-OSS), and the deterministic keyword evaluator. All main-text accuracy numbers are independently reproducible. MIMIC-IV source notes require separate PhysioNet credentialed access. Full system code will be released upon acceptance.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details necessary to understand the results?

   Answer: **Yes.**

   Justification: Section 4 specifies benchmark design, question counts, condition definitions, patient selection criteria, and evaluation methodology. Appendix Q provides model versions and computational requirements.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly?

   Answer: **Yes.**

   Justification: Table 6 reports BCa bootstrap 95% CIs for all conditions. Per-tier results (Figure 3) are clearly labeled as point estimates without per-tier CIs due to small tier-level sample sizes ($n = 2$ patients per tier). The overall B2$\rightarrow$B3 CI crossing zero is explicitly noted.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources needed to reproduce the experiments?

   Answer: **Yes.**

   Justification: Appendix Q reports computational requirements: ClinicalBench Opus $\approx$22 min per condition (API), MedGemma $\approx$3.6 hours (single GPU), SliceBench $\approx$2 hours (API).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

Answer: **Yes.**

Justification: The study uses de-identified clinical data under an approved data use agreement (Appendix Z). No patient re-identification was attempted. The system is designed for clinical decision *support*, not autonomous decision-making.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work?

    Answer: **Yes.**

    Justification: Section 6 discusses the potential for improved clinical decision support (positive) and risks of over-reliance on automated systems, context dilution, and the need for physician oversight (negative). Appendix Z addresses data ethics.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models?

    Answer: **Yes.**

    Justification: The system operates on de-identified data under PhysioNet DUA. It is designed as a decision support tool requiring physician oversight. Code release will include usage guidelines emphasizing the system is not validated for autonomous clinical use.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets used in the paper properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: **Yes.**

    Justification: MIMIC-IV is cited [34] and used under PhysioNet Credentialed Health Data Use Agreement. All benchmark frameworks and models are cited. OMOP CDM is an open standard.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: **Yes.**

    Justification: ClinicalBench and SliceBench are new benchmarks fully documented in Section 4, with question generation procedures, category definitions (Appendix R), and evaluation protocols. Templates and scoring scripts will accompany the code release.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots?

    Answer: **N/A.**

    Justification: No crowdsourcing was used. Physician adjudication (Appendix O) involves expert review by co-authors, not human subjects research.

15. **IRB approvals or equivalent**

Question: Does the paper describe potential risks and does it include the IRB approval number if applicable?

Answer: **N/A.**

Justification: MIMIC-IV is a de-identified public dataset; use under PhysioNet DUA does not require separate IRB approval per PhysioNet policy.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods?

Answer: **Yes.**

Justification: LLMs are core experimental components, not authoring aids. Claude Opus 4.6 is the primary answering model for ClinicalBench (Section 5.1); Claude Sonnet 4.5 answers SliceBench questions while Opus 4.6 judges (Section 5.4); MedGemma 27B provides cross-model validation (Section 5.3). Model versions, temperatures, and inference details are in Appendix Q.